

## RKDF University Bhopal

### Faculty Profile

| Basic Information                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                 |                                            |                    |
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| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | DR. MEENAKSHI SAMARTHA                          |                                            |                    |
| Date of Birth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 31/10/1977                                      |                                            |                    |
| Designation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Professor                                       |                                            |                    |
| Department                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Zoology                                         |                                            |                    |
| Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 15 years after (Research+Teaching)              |                                            |                    |
| Email ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | drmeenakshisamartha@gmail.com                   |                                            |                    |
| Contact No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 9685362966                                      |                                            |                    |
| Educational Qualification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                 |                                            |                    |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Year                                            | University                                 |                    |
| UG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2000                                            | University of Rajasthan, Jaipur, Rajasthan |                    |
| PG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2002                                            | University of Rajasthan, Jaipur, Rajasthan |                    |
| M.Phil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -                                               | -                                          |                    |
| Ph.D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2006                                            | University of Rajasthan, Jaipur, Rajasthan |                    |
| Post Doctorate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | SRF (CSIR)<br>PDF (CPRI, MNRE, Rensselaer, USA) | 2006-2007<br>2021-2022                     |                    |
| NET Qualified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -                                               | -                                          |                    |
| Publications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                 |                                            |                    |
| No. of Research Papers Published (Attach Annexure of Research papers published)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 27                                              |                                            |                    |
| No. of Books Published                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 01                                              |                                            |                    |
| Books Chapters Published (Provide print of 1 <sup>st</sup> Page of Book)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 04                                              |                                            |                    |
| Research Scholars Supervised                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                 |                                            |                    |
| Research Program                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Award                                           | Under Supervision                          | Name of University |
| Ph.D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 02                                              | Dr Meenakshi Samartha                      | RKDF University    |
| M.Phil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                 | -                                          | -                  |
| PG Thesis/Dissertation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 04                                              | Dr Meenakshi Samartha                      | RKDF University    |
| Area of Expertise (100 words)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                 |                                            |                    |
| <p>Life Sciences , (Zoology) specialization in cancer biology,</p> <ul style="list-style-type: none"> <li>Biochemical estimation in blood and liver (LPO, GSH, SGOT, Protein, Fats and carbohydrates) using spectrophotometer.</li> <li>Cytogenetic studies: <b>Chromosomal</b> aberrations, <b>Micronuclei</b> preparations.</li> <li>Handling caring and drug administration to the mice.</li> <li>Maintaining animal house.</li> <li>Preparation of plant extract with the help of Soxhlet apparatus.</li> <li>Tumor models : 1. <b>Two stage mouse skin carcinogenesis</b><br/>2. <b>Lung tumor model system (Yun's Model)</b></li> <li>Various instruments &amp; techniques routinely used in hematological studies like Microscope, Centrifuge, Spectrophotometer, Autoclave, Water bath, BOD incubator, Flow cytometer, Stain preparations, etc.</li> <li>Evaluation of antioxidant activity (Radical Scavenging Activity)<br/><b>(DPPH and ABTS assays)</b></li> <li><b>Blood Culture</b> Techniques</li> <li><b>Histopathological Preparations:</b> Tissue fixation, block making, cutting and staining, observations using light microscope and microphotography.</li> <li><b>Mushroom Cultivation</b> technique</li> </ul> |                                                 |                                            |                    |

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| <ul style="list-style-type: none"><li>• <b>Biofuel Production</b> through Algae Route using Carbon Capture Pilot Plant</li><li>• <b>Fish Biology</b></li><li>• Teaching Zoology, Biochemistry, Biotechnology, Food Science and Nutrition, environmental studies, Microbiology UG and PG students with practical studies.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                |
| <b>Award and Achievement</b>                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                |
| <b>Name of Award</b>                                                                                                                                                                                                                                                                                                               | <b>Description (With certified of Copy award)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                |
| <b>National</b>                                                                                                                                                                                                                                                                                                                    | 1. Research Excellence award (September, 2022) RKDF University, Bhopal.<br><br>2. Award of Excellence in Education, Research and Innovation (Nov-2022) by Resinnovative Science, MPCST, vigyan Bhawan, Bhopal (MP)                                                                                                                                                                                                                                                                                                          |                                                                                                                                |
| <b>International</b>                                                                                                                                                                                                                                                                                                               | 3. <b>YRSA Delegate-</b> Awarded “ <b>Young Radiation Scientists in Asia</b> ” to participate in the “The 1 <sup>st</sup> Asian Congress of Radiation Research (ACRR),” Hiroshima-Japan (2005).<br><br>4. Provided with whole staying expenses by the Symposium Organization (The Korean Society of Ginseng) for the <b>9<sup>th</sup> International Symposium of Ginseng</b> , South Korea.<br><br>5. Best Paper Award in ICSEEGT-2022, International Conference on sustainable Energy Environment and Green technologies. |                                                                                                                                |
| <b>Conference/Seminar/Workshops/FDP</b>                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                |
| <b>Description</b>                                                                                                                                                                                                                                                                                                                 | <b>34</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                |
| <b>Conference/Seminar attended/ organized</b>                                                                                                                                                                                                                                                                                      | <b>28</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                |
| <b>Work shop attended/ organized</b>                                                                                                                                                                                                                                                                                               | <b>03</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                |
| <b>FDP Attended/ organized</b>                                                                                                                                                                                                                                                                                                     | <b>03</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                |
| <b>Research Project</b>                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                |
| <b>Name of Project</b>                                                                                                                                                                                                                                                                                                             | <b>Funding Agencies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Amount</b>                                                                                                                  |
| 1. Skin carcinogenesis induced by DMBA and Croton oil and its modulation in Swiss albino mice                                                                                                                                                                                                                                      | Department of Pharmaceuticals and Health Foods, LOTTE R & D Centre, Seoul, South Korea                                                                                                                                                                                                                                                                                                                                                                                                                                      | Ginseng fractionated products EFLA400, Rg3 and RP1 Drugs provided for chemopreventive studies on Swiss albino mice. (PhD work) |
| 2. Antioxidant, Anticancer and Antimutagenic activity of certain phytochemicals.                                                                                                                                                                                                                                                   | CSIR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Fellowship 12,000 per month plus 20,000/- contingency per year                                                                 |
| 3. Environmental pollution in reference of Kanha Reserve Forest due to tourism, social activities and accidental incidents.                                                                                                                                                                                                        | School of interdisciplinary and trans-disciplinary Studies (SOITS), IGNOU, New Delhi.                                                                                                                                                                                                                                                                                                                                                                                                                                       | Nil                                                                                                                            |
| 4. Biofuel Production through algae route using carbon captures pilot plant.                                                                                                                                                                                                                                                       | MNRE, CPRI and Rensselaer, USA. With RKDF equity                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Part of project as a Post doc fellow                                                                                           |

• **Any other Achievement**

- Project approved by ICMR in 2014-15 entitled “Development and characterization of oral cancer stem cells”.
- **Guest Talk**

1. **Guest Speaker** for one day International Webinar on “**Impact of Pandemic on Agri Education and Research**” organized by School of Agriculture, Eklavya University, Damoh (M.P.) on 15<sup>th</sup> July, 2021. Topic of guest talk :**Impact of Covid-19 on Research work .**
2. **Guest Speaker** in 16<sup>th</sup> India-Japan Bilateral Virtual Conference (BICON-2021) on “**Recreating Higher Education in the Post-Pandemic World**” to be organized from 14<sup>th</sup> December 2021 to 18<sup>th</sup> December 2021. Topic: “**Virtual Laboratory Work and Its Challenges after COVID 19**”.
3. Talk on **Role of Indian Knowledge system in the field of Zoology** July 2024

- **Evaluation of PhD thesis:**

1. **External Examiner for Final Viva Voce of PhD thesis submitted by Ms. Anjali Bhargawa enrollment no: AU172265.** in Department of Life Science at Rabindranath Tagore University, Bhopal 22<sup>nd</sup> December 2022.
2. Thesis evaluated PhD scholar of **Sant Gadge Baba Amravati University** entitled **Ecology and Diversity of Beetles in Pench, Maharashtra**. September 2024.
3. Thesis evaluated PhD scholar of School of Basic and Applied Sciences **Nirwan University, Jaipur, Rajasthan** entitled “**Study on physical properties of Ayurvedic Nanocrystalline bhasma by employing modern scientific tools and approach for cancer treatment**” February 2025

- **Session Chair**

1. International Conference on **Fight Against Cancer (FACCON)** , 5<sup>th</sup> to 7<sup>th</sup> October Chatrapati Shahuji Maharaj University, Kanpur India. Participated as **Programme Director of the scientific session** on October 7, 2017.